

Benjamin H. Silver

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EDUCATION

Tufts University | Medford, MA

Expected May 2028

Bachelor of Arts – Computer Science | Focus in Biological Sciences

- GPA: 3.9/4.0 | Honors: Dean's List (All Semesters)
- Relevant Coursework: General Chemistry with Lab, Cells & Organisms with Lab, Data Structures, Web Programming, Linear Algebra, Machine Structure & Assembly Language Programming, Machine Learning

RESEARCH EXPERIENCE

Undergraduate Researcher | MGH Vaccine & Immunotherapy Center (VIC), Boston, MA

Jul. 2026 – Present

- Engineered a mass cytometry (CyTOF) pipeline in Python to profile immune cell populations across vaccine stimulations, processing 50+ flow cytometry parameters from a dataset with >1 million entries.
- Implemented data preprocessing workflows including FCS file parsing, arcsinh transformation, and dimensionality reduction (PCA, UMAP).
- Augmented existing clustering algorithms using biological cell population validation metrics and graphical analysis to improve cell population assignments and inform immunological interpretation.

Undergraduate Research Assistant | Sonkusale Research Labs, Tufts University

Feb. 2026 – May 2026

- Developed and validated fabrication protocols for flexible electrode arrays, conducting rigorous signal characterization and benchtop testing to optimize biosensor performance across physiological conditions relevant to wearable point-of-care diagnostics.

PROJECTS

AI Allergen Safety App

May 2026

- Developed a full-stack web application (Python, Flask, Claude Vision API, Google Maps API) to analyze food images, detect allergens and deliver risk-level warnings for travelers with severe dietary restrictions.
- Integrated a live map to locate nearby pharmacies and hospitals stocking epinephrine auto-injectors, enabling emergency response abroad, addressing significant health equity gaps in emergency preparedness.

CraveCast | Machine Learning Addiction Aid App

Sep. 2026

- Architected craving prediction tool for addiction recovery, combining peer-reviewed research (ADARP study) with personalized ML modeling (kernel density estimation) to provide a warning system for high-risk periods.
- Single-file HTML app deployed on GitHub Pages reaching online recovery communities and works fully offline to prioritize user privacy.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Research Computing Student Specialist | Tufts Technology Services

Apr. 2025 – Present

- Built web automation tools in HTML and JavaScript for research storage management, supporting optimized quota monitoring and resource allocation across 500+ daily users.

General Body Member | Tufts Students for the Exploration and Development of Space

Sep. 2025 – Present

- Designed and fabricated L1 certification rocket using OnShape (CAD) and OpenRocket simulation; executed laser cutting and assembly for launch.
- Developed proficiency in rocket aerodynamics, stability analysis (CP/CG optimization), nosecone design, and solid motor propulsion systems.

Treasurer & Underclassman Rep. | Tufts National Student Data Corps

Sep. 2025 – May 2026

- Led workshops on Python data analysis (NumPy, Matplotlib, Pandas) for 30+ students using Jupyter Notebooks, with applied examples in scientific computing and data manipulation.

SKILLS & CERTIFICATIONS

Technical: Python, C/C++, R, HTML, JavaScript, SQL, React.js, FastAPI, PyTorch, NumPy, Scikit-learn

Biomedical Data Science: CyTOF Analysis, single-cell analysis, OMIQ, FlowSOM, Clustering Algorithms, single-cell RNA-seq, dimensionality reduction (PCA, UMAP, t-SNE), biological data visualization

Interests: Mountain Biking and Bike Maintenance, Leatherworking, Taiwanese Food, Knife Sharpening